

AI Research Cheat Sheet

Reference Guide

Ultimate AI Research Toolkit 2026 - Premium Edition

Designed for PhD students, professors, graduate researchers, engineers, and data scientists.

[Idea]	Research prompts and ideation systems
[Review]	Literature synthesis and gap detection
[Write]	Paper writing and revision workflows
[Plan]	Publication and productivity operations

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Introduction

A fast-reference sheet for reliable AI prompting in technical and academic contexts.

Prompt Patterns

- Context -> Objective -> Constraints -> Output format -> Quality checks.
- Always require assumptions and confidence limits in analytical outputs.

Validation Rules

- Cross-verify critical claims against cited sources.
- Use at least one adversarial prompt to detect weak reasoning.

Failure Modes

- Overconfident unsupported claims.
- Ignoring boundary conditions and resource constraints.

Summary

Use this sheet before each AI session to improve output reliability and consistency.